# American Computer Science League

2021-2022 ● Contest 3: Short Problems ● Junior Division

**1. Boolean Algebra**

Which ordered pair makes the following Boolean expression TRUE?

$$\overline{A}\,(AB + \overline{B})$$

A. (0, 0)
B. (0, 1)
C. (1, 0)
D. (1, 1)
E. None of the above

**2. Boolean Algebra**

Simplify the following Boolean expression:

$$A\,(\overline{B} + \overline{C}) + \overline{A}\,B + \overline{B\,C}$$

A. $\overline{A} + B + \overline{C}$
B. $B + \overline{A}\,C + \overline{C}$
C. $A\overline{B} + B + \overline{C}$
D. $\overline{B} + \overline{C} + \overline{A}\,B$
E. $C + \overline{A} + \overline{B}$

**3. Data Structures**

What nodes (listed left to right) are at a depth of 3 in the binary search tree for:

TOKYOOLYMPICS

A. S O I
B. C O L
C. I P S
D. K P Y
E. I O S

**4. Data Structures**

Given an initially empty queue and the following commands on the queue, what item will be popped next?

PUSH("A"), PUSH("Q"), PUSH("U"), X=POP( ),
PUSH("A"), PUSH("R"), X=POP( ), X=POP( ),
PUSH("I"), X=POP( ), PUSH("U"), PUSH("S")

A. A
B. I
C. R
D. U
E. S

**5. What Does This Program Do? (Arrays)**

After the following program is executed, how many elements in the array are not zero?

```
for i = 0 to 3
    for j = 0 to 3
        a(i,j) = 2 * i  +  3 * j
    next j
next i
for i = 0 to 3
     for j = 0 to 3
         if a(i,j) >= 10 then
             a(i,j) = a(i,j) − 10
         end if
         if a(i,j) < 6 then
             a(i,j) = 0
         end if
      next j
next i
```

A. 0
B. 4
C. 6
D. 10
E. 16

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**1. 布尔代数**

哪个有序对可以使得下方的布尔表达式为真？

$$\overline{A}\,(AB + \overline{B})$$

A. (0, 0)
B. (0, 1)
C. (1, 0)
D. (1, 1)
E. 以上均不是

**2. 布尔代数**

化简以下布尔表达式：

$$A\,(\overline{B} + \overline{C}) + \overline{A}\,B + \overline{B\,C}$$

A. $\overline{A} + B + \overline{C}$
B. $B + \overline{A}\,C + \overline{C}$
C. $A\overline{B} + B + \overline{C}$
D. $\overline{B} + \overline{C} + \overline{A}\,B$
E. $C + \overline{A} + \overline{B}$

**3. 数据结构**

在**下述**二叉搜索树中，深度为 3 的节点（从左到右列出）**有哪些**：

TOKYOOLYMPICS

A. S O I
B. C O L
C. I P S
D. K P Y
E. I O S

**4. 数据结构**

**给出一个初始为空的队列，并在该队列上执行以下命令，请问接下来将要出队的元素是什么**?

PUSH(“A”), PUSH(“Q”), PUSH(“U”), X=POP( ),
PUSH(“A”), PUSH(“R”), X=POP( ), X=POP( ),
PUSH(“I”), X=POP( ), PUSH(“U”), PUSH(“S”)

A. A
B. I
C. R
D. U
E. S

**5. 此程序的作用是什么？ (数组)**

**下述**程序**执行完毕**后，**数组**中有多少个非零元素？

```
for i = 0 to 3
    for j = 0 to 3
        a(i,j) = 2 * i  +  3 * j
    next j
next i
for i = 0 to 3
     for j = 0 to 3
         if a(i,j) >= 10 then
             a(i,j) = a(i,j) − 10
         end if
         if a(i,j) < 6 then
             a(i,j) = 0
         end if
      next j
next i
```

A. 0
B. 4
C. 6
D. 10
E. 16